INDUSTRIAL DRIVES

The following are the general conditions that may limit (affect) the choice of an electric motor

(a) Supply Available

- If a.c, the voltage, frequency and number of phases of the motor must suit (match with) the supply.
- If d.c, costly rectifying and control equipment may be required.
- If a new motor is to be installed consider if there is sufficient spare capacity at the supply intake point to cater for the additional load.
- Power factor of the load, should also be considered as it affect the tariff charges.

(b) Starting and Control

- Limit set to the starting current will rule out some types of motors and starters (ie supply authority will not allow the use of the direct-on-line of a cage induction motor above a certain rating).
- The frequency of starting will influence the type of starter. ie where the motor is continually started and stopped, or otherwise, the starting equipment must allow (be suitable) for this purpose.
- If speed control is required some special kind of motor (a.c or d.c) may be required with electronic control, which may be costly.
- Reversal of direction if needed should be suitable.

(c) Enclosures:

The enclosure will depend upon the atmosphere in which the motor is working. ie

- clean air
- moist atmosphere
- presence of dust
- chemical vapour
- hot atmosphere
- flammable gas/vapour (explosive situation)

All these conditions will determine the type of motor enclosure.

MOTOR ENCLOSURES

The type of casing or motor enclosure will greatly depend upon the conditions under which the motor has to function. Such conditions includes

- Clean air
- Chemical vapours
- Moist or watery atmosphere
- Presence of dust
- hot atmosphere
- flammable or explosive situations
- etc.

Purpose of an enclosure.

(i) — to prevent the user from making an accidental contact with the live or rotating part of the machine, and

(ii) — to protect the motor from damage due to the conditions in which it functions. i.e dust, dampness, explosive situation etc.

In dry dust free atmospheres, open machines (motors) are used to drive machineries. These are sometimes fenced off to prevent any unauthorised person or livestock from coming into contact with conductors or moving parts. However this method is obviously unsuitable unless the conditions are so ideal, and no work is likely to be carried out near the motor.

Types of motor enclosures.

(1) Open machine.

— The ends of this machine are completely

open to allow free ventilation for circulating air, over and through the windings.
— Air is drawn through the motor by a fan attached to the motor shaft.
— This type is only suitable for perfectly clean and dry atmosphere. It may be used in generator and motor room in which only authorised persons are allowed to enter.

(2) Screened or Screen-protected.

— The motor is enclosed in a steel casing and the shaft runs on bearings housed in the end covers of the casing. The end covers are fitted with metallic screens and an internal fan is provided to drive the cooling air through the motor.
— All the ventilations in the frame are protected by wire screen to avoid contact with life parts and the rotating parts of the motor. Other types of screens may be, wire mesh, expanded metal, perforated metal or similar covers.

This enclosure is suitable for motors used in dry, dust free conditions and is the most common type of enclosure in use.

(3) Totally enclosed.

This enclosure has no air inlet and for that cooling is achieved by means of radiation and conduction from through the surface of the motor. An internal fan is provided and the casing is ribbed to increase the cooling surface. This means that the internal air has no connection with the external air and the internal circulation is maintained by the fan.

This motor runs hotter than any other type and therefore, a high-class insulation material such as fibre glass is often used to withstand the higher temperatures.

Since dirt and moisture are excluded, this motor (enclosure ~~enclosure~~) is suitable in corrossive atmospheres, but care must be taken to ensure that the ribs are not clogged up to prevent overheating. It can also be used under the following conditions;

- weather proof
- water tight machine
- Submersible machine.

(4) Flame proof enclosure

– Similar to totally enclosed but the machine is physically larger and more robust. The motor is totally enclosed in a flameproof enclosure and the enclosure should correspond with the flame proof enclosure specifications and regulations. ie should be able to withstand an internal explosion without allowing the passage of a dangerous flame which could ignite the external atmosphere.

These machines are applicable in mining and oil industries and other situations where explosive atmosphere is encountered (ie presence of flammable dust or liquids, gas or vapour).

NB Great care is required in the maintenance of a flame proof (enclosure) machine.

(5) Drip Proof enclosure.

~~– Simlar Similar to the screened type but the screens are fitted~~ (openings)

Similar to the screened type but the ventilating openings are so constructed that any moisture or dirt falling vertically cannot enter the machine. These machines are suitable for damp situations but they are not water-proof.

The enclosure will only prevent the ingress of dripping liquid or dirt particles but not flowing water into the motor. This enclosure is also Not suitable for dusty conditions as particles of dust may block the screen leading to overheating.

c) Pipe or duct Ventilated.

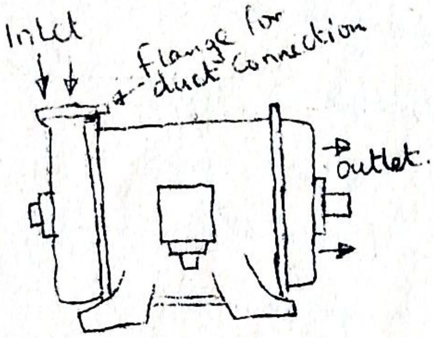
If the air inside the room in which the motor is situated is not suitable for cooling the motor (ie too hot, damp or dusty, with chemical fumes and vapours etc), cool and clean air is drawn in from outside. This is done by means of a pipe or duct connected to the ventilation inlet of the motor. The outlet pipe (duct) is also provided so that the air ~~so that air~~ flowing in, is circulated back to the outside by the means of an internal fan.

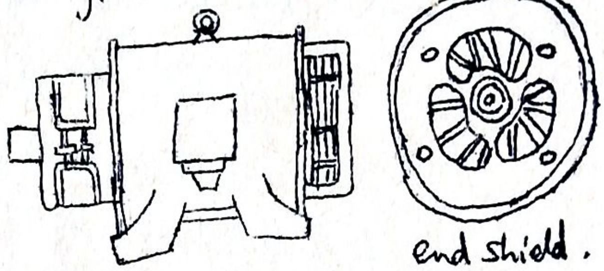
This machine is widely used in situation where it is very dusty or otherwise as earlier mentioned, like in woodworking shops.

MOTOR ENCLOSURES

Note: Except the open Machine all types of motor enclosures must give protection to safeguard against accidental or inadvertent contact with the internal moving or Live parts.

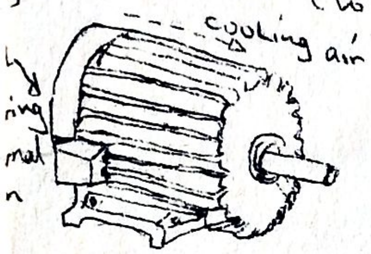
elow are Some of the Various types of Motor enclosures:

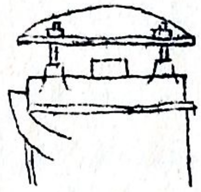




(b) Screen Protected.

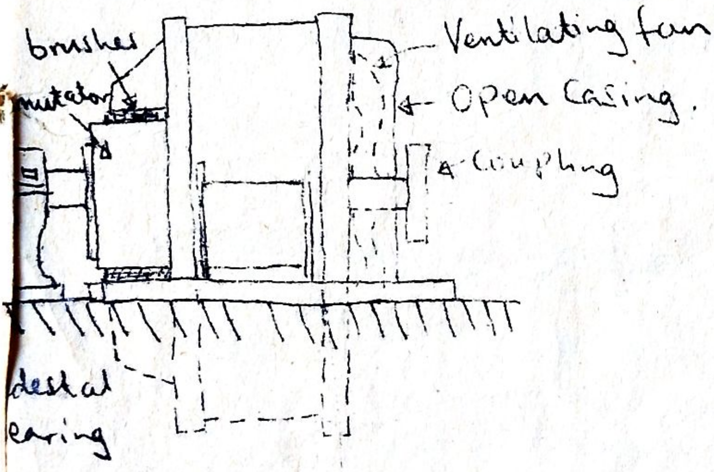
y Duct Ventilated (duct inlet, outlet to room)





(d) Drip-proof protection.

Totally enclosed Machine



Open Machine with Pedestal bearing.

ASSIGNMENT / Revision Questions

A 15hp motor driving an air compressor is located n a Workshop in dirty Surroundings, and supplies an adjacent air ~~cleaner~~ receiver, Under the Control of a Pressure

INDUSTRIAL APPLICATION OF ELECTRIC MOTORS.

Advantages of Electric Drive as Compared with mech. drives.

i) – simple in construction and less maintenance cost.

(ii) – Speed Control is easy and smooth.

(iii) – neat, clean and free from any smoke or flue gases.

(iv) – Can be installed at any desired convenient place, thus affording more flexibility in the layout

(V) – Can be remotely controlled

(VI) – being Compact, requires less space.

(Vii) – Can be started immediately without any loss of time.

(viii) – has Comparatively longer life.

Disadvantages of electric drives.

(i) – Comes to stop as soon as there is failure of electric supply

(ii) – Can NOT be used in places without electric supply.

NB The above mentioned, two demerits can be over-come by installing a diesel-driven generator to serve in absence of electric power or when normal electric supply fails.

CLASSIFICATION OF ELECTRIC DRIVES

Three Categories: – group drive.
– Individual drive
– Multi motor drive.

(a) Group drive: a single motor drives a number of machines through belts from a common shaft – also called "shaft drive"

(b) Individual drive - each m/c driven by its own separate motor with help of gears, pulley etc.

(c) Multi motor drives; - Separate motors provided ~~to~~ for actuating different parts of the driven mechanism, ie in travelling cranes, three motors are used; one for hoisting, another for long travel motion and the third for cross travel motion

- Commonly used in paper mills, rolling mills, rotary printing presses and metal working machines. etc.

Advantages Of Group drive:

(i) saving in initial cost as one big motor cost much less than several small motor to do the same duty (ie One 150 kW motor compared to ten 15kW motor.)

(ii)

(iii)

(iv)

(v)

(vi)

(vii)

Disadvantage of Individual drive.

- Initial cost is high

SELECTION OF A MOTOR.

The following are the factors to be considered.

(a) Electrical characteristics.
- starting characteristics
- speed control
- running characteristics
- braking.

(b) Mechanical Considerations.
- type of enclosure.
- method of power transmission
- noise level
- type of bearing
- type of cooling.

(c) Size and rating of motor.
- requirements for continuous, intermittent or Variable load cycle
- overload capacity.

(d) Cost.
- Capital cost
- Running cost.